

Objects365: A Large-scale, High-quality Dataset for Object Detection

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Abstract

In this paper, we introduce a new large-scale object detection dataset, *Objects365*, which has 365 object categories over 600K training images. More than 10 million, high-quality bounding boxes are manually labeled through a three-step, carefully designed annotation pipeline. It is the largest object detection dataset (with full annotation) so far and establishes a more challenging benchmark for the community. *Objects365* can serve as a better feature learning dataset for localization-sensitive tasks like object detection and semantic segmentation. The *Objects365* pre-trained models significantly outperform ImageNet pre-trained models with 5.6 points gain (42 vs 36.4) based on the standard setting of 90K iterations on COCO benchmark. Even compared with much long training time like 540K iterations, our *Objects365* pretrained model with 90K iterations still have 2.7 points gain (42 vs 39.3). Meanwhile, the finetuning time can be greatly reduced (up to 10 times) when reaching the same accuracy. Better generalization ability of *Object365* has also been verified on *CityPersons*, *VOC* segmentation, and *ADE* tasks. The dataset as well as the pretrained-models have been released at www.objects365.org.

1. Introduction

Object detection is a fundamental task in computer vision. PASCAL VOC [8] and COCO [24], have contributed greatly to rapid advances of object detection. From traditional approaches like DPM [9] to deep-learning based approaches like R-CNN [13] and FPN [22], the above two datasets serve as “golden” benchmarks to evaluate algorithms and boost research progresses. In this paper, we move a step further to introduce a new large-scale, high-quality object detection dataset, *Objects365*, which focuses on three aspects: *scale*, *quality*, and *generalization*.

Scale. *Objects365* is significantly larger than the exist-

*indicates equal contribution.

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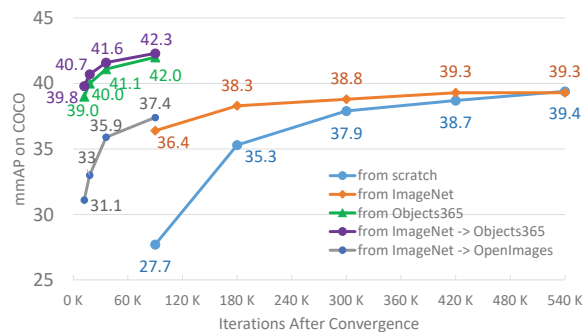


Figure 1. Results of finetuning COCO from ImageNet vs *Objects365*. With a small number of iterations, 90K, for training, our *Objects365* pre-trained model (green curve) can significantly outperform the algorithm finetuned from ImageNet (orange curve), even with much longer training time, e.g., 540K iterations. All results use a FPN based Resnet50 backbone.

ing object detection benchmarks like PASCAL and COCO. It contains 365 categories, 638K images, and 10,101K bounding boxes. We compare our dataset with existing object detection benchmarks with full annotations in Table 1. Our dataset contains 5× more images, 4× more categories, and 10× more boxes than COCO [24]. It can serve as a more challenging benchmark for the detection community.

Quality. In addition to the size, annotation quality is of great importance when building a dataset. To ensure quality, we divide the annotation pipeline into three steps, which can significantly reduce the job requirement for the annotators. Besides the annotators, we also include inspectors and examiners to review the quality of the annotations. To reduce ambiguities during the annotation process, we apply two consistency rules. This annotation pipeline ensures that we obtain high-quality annotation with high efficiency.

Generalization. The feature learned from *Objects365* is superior for many localization-sensitive tasks like object detection and semantic segmentation. Conventionally, ImageNet [5] pre-trained basenets like Resnet [17] are widely employed as a backbone for the object de-

Objects365：一个用于目标检测的大规模高质量数据集

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摘要

In this paper, we introduce a new large-scale object detection dataset, *Objects365*, which has 365 object categories over 600K training images. More than 10 million, high-quality bounding boxes are manually labeled through a three-step, carefully designed annotation pipeline. It is the largest object detection dataset (with full annotation) so far and establishes a more challenging benchmark for the community. *Objects365* can serve as a better feature learning dataset for localization-sensitive tasks like object detection and semantic segmentation. The *Objects365* pre-trained models significantly outperform ImageNet pre-trained models with 5.6 points gain (42 vs 36.4) based on the standard setting of 90K iterations on COCO benchmark. Even compared with much long training time like 540K iterations, our *Objects365* pretrained model with 90K iterations still have 2.7 points gain (42 vs 39.3). Meanwhile, the finetuning time can be greatly reduced (up to 10 times) when reaching the same accuracy. Better generalization ability of *Object365* has also been verified on CityPersons, VOC segmentation, and ADE tasks. The dataset as well as the pretrained-models have been released at www.objects365.org.

1. 引言

目标检测是计算机视觉中的一项基础任务。PASCAL VOC [8] 和 COCO [24] 极大地推动了目标检测的快速发展。从 DPM [9] 等传统方法到 R-CNN [13] 和 FPN [22] 等基于深度学习的方法，上述两个数据集都作为评估算法和推动研究进展的“黄金”基准。在本文中，我们更进一步，引入了一个新的大规模、高质量目标检测数据集 *Objects365*，其重点关注三个方面：*scale*、*quality* 和 *generalization*。

Scale *Objects365* 的规模显著大于现有的

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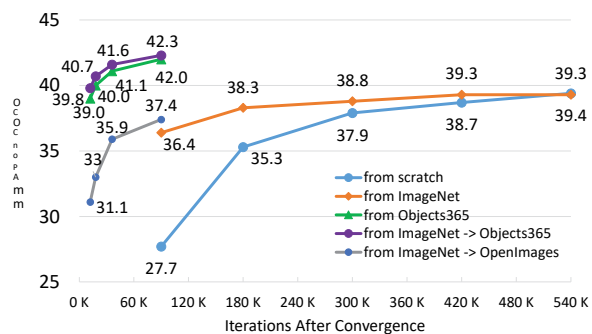


图1. 基于ImageNet与Objects365对COCO进行微调的结果对比。在仅使用9万次迭代的小规模训练条件下，我们基于Object s365的预训练模型（绿色曲线）显著超越了基于ImageNet预训练进行微调的算法（橙色曲线），后者即使经过更长时间的训练（例如54万次迭代）仍未能取得优势。所有实验结果均采用基于FPN的Resnet50骨干网络。

在PASCAL和COCO等目标检测基准测试中。它包含365个类别、638K张图像和10,101K个边界框。我们在表1中将我们的数据集与现有全标注目标检测基准进行了比较。相比COCO[24]，我们的数据集图像数量多5×倍、类别多4×倍、边界框多10×倍。它可以为检测领域提供一个更具挑战性的基准。

*Quality*除了数据集的规模，标注质量在构建数据集时也至关重要。为确保质量，我们将标注流程分为三个步骤，这能显著降低对标注人员的工作要求。除了标注人员，我们还引入了检查员和审核员来审查标注质量。为减少标注过程中的歧义，我们应用了两条一致性规则。这一标注流程确保我们能够高效地获得高质量的标注结果。

*Generalization*从Objects365学习到的特征在许多对定位敏感的任务中表现更优，例如目标检测和语义分割。传统上，像ResNet [17]这样基于ImageNet [5]预训练的基网络被广泛用作目标检测的骨干网络。

tection/segmentation algorithms. However, there are two issues related to ImageNet pretraining. On one hand, the feature learned on ImageNet classification task is less localization-sensitive. On the other hand, only the backbone part are pre-trained but the head part is initialized with the random weights.

Our Objects365 dataset directly addresses the above two issues and provides a better alternative for feature learning. As shown in Figure 1, the Objects365 pre-trained features can significantly outperform the counterparts based on ImageNet, even the one with sufficient longer training time (540K iterations) as discussed in [15]. Moreover, using Objects365 feature, we can obtain comparable results with one order-of-magnitude less training time.

2. Related Work

General Object Detection Object detection task has been a fundamental research topic for a while. DPM [9] is one of the most famous object detection algorithms to be used before the introduction of deep learning techniques. R-CNN [13] is one of the first work to integrate the convolutional neural network for object detection. Later, with the rapid development of convolutional neural network, most object detection algorithms started to utilize deep learning techniques. Roughly, we can divide the existing object detection algorithms into two categories: single-stage detector and two-stage detector. The main difference lies on whether to pool the feature maps for a second stage. SSD [25], DSSD [10] and YOLO series [27, 28, 29] are some widely used single-stage detector with efficient speed. RetinaNet [23] is introduced with strong performance even compared with the two-stage detector. For the two-stage detector, the early work like Fast R-CNN [12], Faster R-CNN [30], R-FCN [3], try to speed up the algorithms. In FPN [22] and Mask R-CNN [16], feature pyramid structure and ROI-Align are proposed to boost the performance. Deformable ConvNet [4, 39], Soft Sampling [34], SNIP [32], SNIPER [33], and Cascade R-CNN [1], DetNet [21] are introduced to further improve the performance.

Large-scale Detection Dataset The large-scale dataset is an important reason for the continuous improvement of the object detection algorithms, especially for deep learning based techniques. From early datasets like ImageNet [5], VOC [8], to the recent benchmarks like COCO [24], they all play an important role in the image classification and object detection community. In Table 1, we give statistics of the existing object detection benchmarks together with our Objects365 benchmark. Our Objects365 dataset has around 60 times images larger than PASCAL VOC and 5 times larger than COCO. Compared with the ImageNet DET dataset [5], our dataset has a larger number of boxes per image, with 15.8 vs 1.1 (2.3 for the Dense set).

Besides the general object detection datasets, there are also a lot of other detection benchmarks like face detection [19, 35], pedestrian detection [7, 36, 31], and human/vehicle detection for the autonomous driving [11, 2], all of which play an important role in the detection community.

3. Objects365 Dataset

In this section, we present the details on the collection, annotation, statistics, and quality of the dataset respectively.

3.1. Data Collection

3.1.1 Data Source

To make the image sources more diverse, we collect images mainly from Flickr¹. All the images conform to licensing for research purposes. Sample images can be found on our website².

3.1.2 Object Categories

Based on the collected images, we first select eleven super-categories which are common and diverse to cover most object instances. They are: *human and related accessories, living room, clothes, kitchen, instrument, transportation, bathroom, electronics, food (vegetables), office supplies, and animal*. Based on the super-categories, we further propose 442 categories which widely exists in our daily lives. As some of the object categories are rarely found, we first annotate all 442 categories in the first 100K images and then select the most frequent 365 object categories as our target objects. Also, to be compatible with the existing object detection benchmarks, the 365 categories include the categories defined in PASCAL VOC [8] and COCO [24] benchmarks.

3.1.3 Non-Iconic Images

As our Objects365 dataset focuses on object detection, we eliminate those images which are only suitable for image classification. For example, the image only contains one object instance around the image center. This filtering process was first adopted in COCO [24].

3.2. Annotation

As there are a large number of images and object categories, a good annotation process is of great importance to ensure high quality and efficiency.

检测/分割算法。然而，与ImageNet预训练相关存在两个问题。一方面，在ImageNet分类任务上学习到的特征对定位敏感度较低。另一方面，只有骨干网络部分进行了预训练，而头部网络部分仍以随机权重初始化。

我们的Objects365数据集直接解决了上述两个问题，并为特征学习提供了更优选择。如图1所示，基于Objects365的预训练特征显著超越了基于ImageNet的对应模型，甚至优于文献[15]中讨论的经过更长时间训练（54万次迭代）的版本。此外，使用Objects365特征时，我们仅需十分之一的训练时间即可获得相当的结果。

2. 相关工作

通用目标检测 目标检测任务长期以来一直是基础课题研究。在深度学习技术引入之前，DPM [9] 是最著名的目标检测算法之一。R-CNN [13] 是首个将卷积神经网络应用于目标检测的开创性工作。随着卷积神经网络的快速发展，多数目标检测算法开始采用深度学习技术。现有目标检测算法大致可分为两类：单阶段检测器与两阶段检测器，其主要区别在于是否对特征图进行二次池化处理。SSD [25]、DSSD [10] 及 YOLO 系列 [27, 28, 29] 是广泛使用的单阶段检测器，具有高效的处理速度。RetinaNet [23] 即使与两阶段检测器相比也展现出强劲性能。在两阶段检测器中，早期研究如 Fast R-CNN [12]、Faster R-CNN [30]、R-FCN [3] 致力于提升算法速度。FPN [22] 与 Mask R-CNN [16] 通过提出特征金字塔结构和 ROI-Align 提升了检测性能。Deformable ConvNet [4, 39]、Soft Sampling [34]、SNIP [32]、SNIPER [33]、Cascade R-CNN [1] 及 DetNet [21] 的提出则进一步推动了性能优化。

大规模检测数据集 大规模数据集是目标检测算法持续改进的重要原因，特别是对于基于深度学习的技术而言。从早期的ImageNet [5]、VOC [8]等数据集，到近期的COCO [24]等基准测试，它们都在图像分类和目标检测领域扮演着重要角色。在表1中，我们统计了现有目标检测基准数据集以及我们的Objects365基准数据。我们的Objects365数据集图像数量约为PASCAL VOC的60倍，是COCO的5倍。与ImageNet DET数据集[5]相比，我们的数据集每张图像包含更多标注框，数量为15.8个（Dense子集为2.3个），而ImageNet DET仅为1.1个。

除了通用的目标检测数据集外，还有许多其他检测基准，如人脸检测[19, 35]、行人检测[7, 36, 31]以及自动驾驶中的人/车辆检测[11, 2]，这些都在检测领域发挥着重要作用。

3. Objects365 数据集

在本节中，我们将分别介绍数据集的收集、标注、统计及质量方面的详细信息。

3.1. 数据收集

3.1.1 数据来源

为了使图像来源更加多样化，我们主要从Flicker¹收集图像。所有图像均符合研究用途的许可要求。示例图像可在我们的网站²上查看。

3.1.2 对象类别

基于收集到的图像，我们首先选取了十一个常见且多样化的超级类别，以覆盖大多数物体实例。它们分别是：*human and related accessories*、*living room*、*clothes*、*kitchen*、*instrument*、*transportation*、*bathroom*、*electronics*、*food (vegetables)*、*office supplies*和*animal*。在这些超级类别的基础上，我们进一步提出了442个广泛存在于日常生活中的类别。由于部分物体类别较为罕见，我们首先在前10万张图像中标注了全部442个类别，随后选取出现频率最高的365个物体类别作为目标对象。同时，为了与现有物体检测基准兼容，这365个类别包含了PASCAL VOC [8] 和 COCO [24] 基准中定义类别。

3.1.3 非图标图像

由于我们的Objects365数据集专注于目标检测，我们剔除了那些仅适用于图像分类的图像。例如，图像中心仅包含一个目标实例。这种过滤流程最初在COCO数据集中被采用[24]。

3.2. 标注

由于图像和物体类别数量庞大，良好的标注流程对于确保高质量和高效率至关重要。

Dataset	Images	Boxes	Categories	Boxes/img	Fully Annotated
Pascal VOC	11.5k	27k	20	2.4	Yes
ImageNet All	477k	534k	200	1.1	Yes
ImageNet Dense	80k	186k	200	2.3	Yes
COCO	123k	896k	80	7.3	Yes
OpenImages	1,515k	14,815k	600	9.8	Partial
Objects365	638k	10,101k	365	15.8	Yes

Table 1. Comparison of the dataset statistics with existing object detection benchmarks. The table includes statistics for training and validation sets.

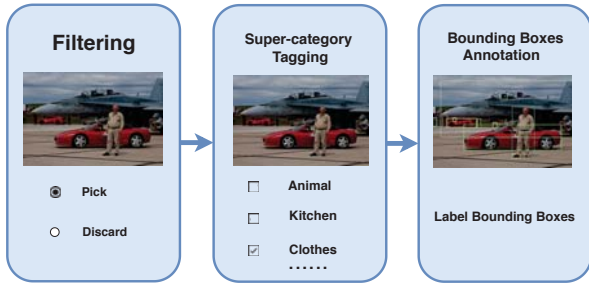


Figure 2. Our annotation pipeline for Objects365.

3.2.1 Annotation Pipeline

It is almost impossible for an annotator to remember and annotate all 365 categories. Also, a small number of images should be rejected due to the iconic images or the images without 365 object categories. Motivated by existing datasets like ImageNet [5] and COCO [24] as well as the discussion of scalable multi-class annotation in [6], we design our annotation pipeline as the following three steps. The first step performs a two-class classification. If the image is non-iconic or contains at least one object instance in the eleven super-categories, it will be passed to the next step. In the second step, the image-level tags with the eleven super-categories will be labeled. An image may be labeled with more than one tag. In the third step, one annotator will be assigned to label the object instances in one specific super-category. All object instances belonging to the super-category should be labeled with a bounding box together with an object name. An illustration of our annotation pipeline is in Figure 2.

Based on the proposed annotation pipeline, each annotator only needs to be familiar with the object categories in one super-category rather than all 365 object categories. This will not only improve annotation efficiency but also boost annotation quality.

3.2.2 Annotation Team

We divide all the team members (hired as vendors) into three groups: annotators, inspectors, examiners. All the images are first annotated by the annotators, and then checked

by the inspectors. By accumulating around 100 labeled (and checked) images as a job, we involve the examiner to further review the quality of the annotation. The job will be rejected if any of the steps exists annotation error.

Annotator The annotators need to perform the annotation which involves all three steps in the annotation pipeline shown in Figure 2. Each annotator will be assigned one and only one annotation task. Before starting the annotation, they should take a course and pass an examination to be qualified for the annotation.

Inspector The work of inspector is to examine all the annotated images labeled by the annotators. If an annotation error is found in one image, the image will be rejected and annotated again by the same annotator. This refining step can greatly prevent the annotator from making the same mistakes in the following annotation process.

Examiner Examiners are usually geographically isolated from the annotators and inspectors to make a fair judgment. The job, which usually contains around 100 images verified by the inspectors, will be reviewed again by the examiner. If an annotation error is found in one image, the whole job which includes the mislabeled image will be rejected. The rejection rate of the jobs is one of the key factors to determine the income for the annotators and inspectors.

3.2.3 Annotation Process Consistency

Due to the large scale of the dataset, there will be a lot of annotators involved in this project. Without consistent definition and rules for the annotation, the labeling for the same image will provide different annotation results. To reduce the ambiguities during the annotation process, we define a number of rules. Two important rules are: classification rule and bounding-box rule.

Classification Rule It defines a clear priority order with **function-first** principle for the ambiguity case in labeling. For instance, in the left Figure 3, the object can be considered as either “tap” or “teapot”. Based on our classification rule, we use the function-first principle and the object will be labeled as “tap” in this case.

¹<https://www.flickr.com/>

²<https://www.objects365.org>

Dataset	Images	Boxes	Categories	Boxes/img	Fully Annotated
Pascal VOC	11.5k	27k	20	2.4	Yes
ImageNet All	477k	534k	200	1.1	Yes
ImageNet Dense	80k	186k	200	2.3	Yes
COCO	123k	896k	80	7.3	Yes
OpenImages	1,515k	14,815k	600	9.8	Partial
Objects365	638k	10,101k	365	15.8	Yes

表1. 数据集统计量与现有目标检测基准的比较。该表格包含训练集和验证集的统计信息。

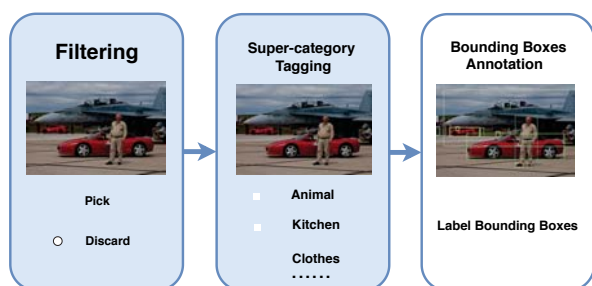


图2. 我们在Objects365数据集上的标注流程。

3.2.1 标注流程

对于标注者来说，记住并标注所有365个类别几乎是不可能的。此外，少量图像因属于标志性图像或不包含365个对象类别而应被剔除。受ImageNet[5]和COCO[24]等现有数据集以及[6]中关于可扩展多类标注讨论的启发，我们将标注流程设计为以下三个步骤。第一步进行二分类判断：若图像为非标志性图像或包含至少一个属于十一个超类别的对象实例，则进入下一步。第二步将标注图像级别的十一个超类别标签，一张图像可能被标注多个标签。第三步中，每位标注者将被分配标注特定超类别中的对象实例，属于该超类别的所有对象实例均需用边界框标注并附上对象名称。我们的标注流程示意图见图2。

基于提出的标注流程，每位标注者只需熟悉一个超类别中的物体类别，而非全部365个物体类别。这不仅会提高标注效率，还能提升标注质量。

3.2.2 标注团队

我们将所有团队成员（以供应商形式雇佣）分为三组：标注员、检查员、审核员。所有图像首先由标注员进行标注，随后由检查员核查。

由检查员进行。通过将大约100张已标注（并检查过）的图像作为一个任务积累起来，我们邀请审查员进一步评估标注的质量。如果任何步骤存在标注错误，该任务将被拒绝。

标注员需要执行标注工作，这涉及图2所示标注流程中的所有三个步骤。每位标注员将被分配且仅分配一项标注任务。在开始标注前，他们必须参加培训并通过考核，以获得标注资格。

检查员的工作是审查所有由标注员标注的图像。如果在一张图像中发现标注错误，该图像将被拒绝，并由同一标注员重新标注。这一精炼步骤能极大地防止标注员在后续标注过程中犯同样的错误。

审查员 审查员通常与标注员和检查员在地理上隔离，以确保做出公正的判断。通常包含约100张由检查员验证过图像的任务，将由审查员再次审核。若在一幅图像中发现标注错误，包含该错误标注图像的整个任务将被拒绝。任务被拒率是决定标注员和检查员收入的关键因素之一。

3.2.3 标注过程一致性

由于数据集规模庞大，本项目将涉及大量标注人员。若缺乏统一的标注定义与规则，同一图像的标注工作将产生不同的注释结果。为减少标注过程中的歧义，我们制定了一系列规则，其中两项重要规则是：分类规则与边界框规则。

分类规则 它通过功能优先原则为标注中的歧义情况定义了明确的优先级顺序。例如，在图3左侧中，该物体可被视为“水龙头”或“茶壶”。根据我们的分类规则，采用功能优先原则，此情况下该物体将被标注为“水龙头”。

¹<https://www.flickr.com/>

²<https://www.objects365.org>



Figure 3. An example showing our function-first consistency rule. In the left figure, the object in the blue bounding-box will be annotated as “tap” rather than “teapot” while the green bounding box will be annotated as “toy” instead of “bear” in the right figure.



Figure 4. An example of our bounding-box consistency rule.

	Train	Validation	Test
Images	600k	38k	100k
Boxes	9623k	479k	1700k

Table 2. The split of our Objects365 benchmark.

Bounding Box Rules Due to the diversity of annotators, the annotation for the bounding-boxes might be inconsistent sometimes. We define the following rule when ambiguities exist for the bounding-box. The annotator is required to cover the largest bounding box which would not lead to the ambiguities of defining the object category. For example, we need to include the decoration part of the clock in the left figure of Figure 4 as the decoration part belongs to the clock and would not lead to misunderstanding of the object category. For the right figure of Figure 4, the annotators are required to label the small bounding box because the external area of the clock will lead to another category as “tower”.

3.3. Statistics

Based on our proposed annotation pipeline, around 740K images are annotated in our Objects365 dataset. A split of train, validation, test set is in Table 2. There are 600K images for training, 38K images for validation, and another 100K images for testing.

To delve into the details of our Objects365, we first provide the statistics of the number of object categories per image. According to Figure 5, Objects365 is more dense and diverse than VOC [8], COCO [24], and OpenImages [20]. Quantitatively, our Objects365 has 5 categories (on aver-

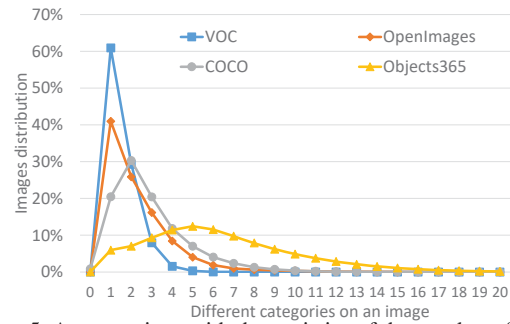


Figure 5. A comparison with the statistics of the number of object categories per image. Objects365 is more dense, with a modal on 5 categories per image, and more diverse, with a more flat curve.

age) per image, which is more dense compared with the other object detection benchmarks like VOC and COCO. Due to the partial annotation used in OpenImages, the category number/image of OpenImages is lower than COCO and Objects365 even though there are 600 categories defined in OpenImages. Moreover, based on the curve in Figure 5, the variance of the number of object categories per image from our Objects365 is significantly larger than the existing benchmarks, which shows the diverse nature of our Objects365 dataset.

For the image resolution, Figure 6 shows that Objects365 has larger and more diverse image resolutions.

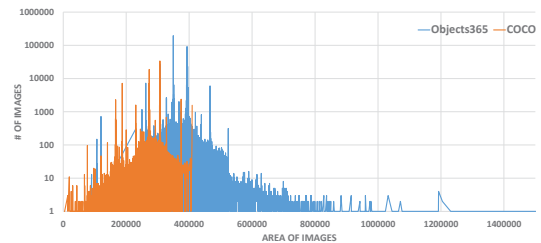


Figure 6. Distribution of the image resolution.

Figure 7 provides a comparison of “effective annotations” with the PASCAL VOC and COCO benchmark. By selecting the COCO 80 categories from Objects365, i.e., ignoring the other object categories, our dataset has more boxes per image, i.e., 9.04 vs 7.34, as well as more object categories per image, 3.16 vs 2.92. A similar conclusion can be obtained by mapping the object categories into the VOC 20 categories. We also compare the effective annotation areas, i.e., the ratio of annotated object areas to the total area of the images. Our Objects365 provides obvious more annotations 63% compared with 57% in COCO and 53% in VOC.



图3. 展示我们“功能优先”一致性规则的示例。左图中，蓝色边界框内的物体将被标注为“水龙头”而非“茶壶”；右图中，绿色边界框内的物体则会被标注为“玩具”而非“泰迪熊”。

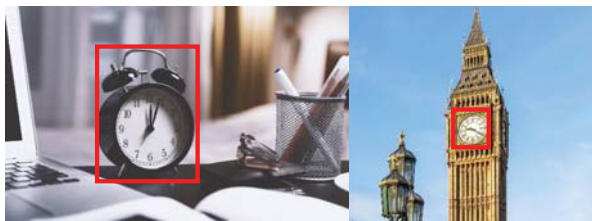


图4. 我们边界框一致性规则的一个示例。

	Train	Validation	Test
Images	600k	38k	100k
Boxes	9623k	479k	1700k

表2. 我们Objects365基准测试的数据划分。

边界框规则 由于标注人员的多样性，边界框的标注有时可能存在不一致。当边界框存在歧义时，我们定义以下规则：标注者需标注**不引起物体类别歧义的最大边界框**。

例如，在图4左图中，我们需要将钟表的装饰部分包含在内，因为装饰部分属于钟表且不会导致对物体类别的误解。而在图4右图中，标注者需标注较小的边界框，因为钟表的外部区域会导致其被归类为“塔”。

3.3. 统计

基于我们提出的标注流程，Objects365数据集中约有74万张图像被标注。训练集、验证集和测试集的划分如表2所示。其中60万张图像用于训练，3.8万张用于验证，另有10万张用于测试。

为了深入了解我们的Objects365数据集，我们首先提供了每张图像中物体类别数量的统计数据。根据图5所示，Objects365比VOC[8]、COCO[24]和OpenImages[20]更密集且更多样化。量化来看，我们的Objects365平均每张图像包含 $\{v^*\}$ 个类别——

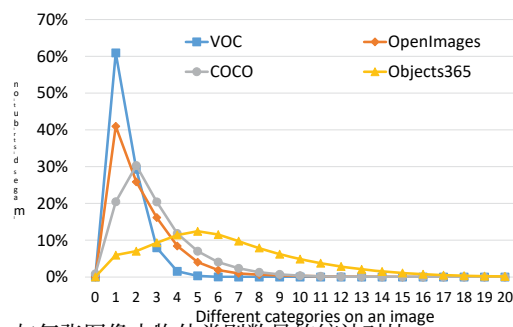


图5. 与每张图像中物体类别数量的统计对比。Objects365数据集更为密集，每张图像的模态为5个类别，且更具多样性，其分布曲线更为平缓。

每张图像类别数（即每张图像的平均标注类别数）相比其他目标检测基准数据集（如VOC和COCO）更为密集。由于OpenImages采用部分标注策略，尽管其定义了600个类别，但其每张图像的类别数仍低于COCO和Objects365。此外，根据图5中的曲线，我们的Objects365数据集中每张图像包含的目标类别数量方差显著大于现有基准数据集，这体现了Objects365数据集的多样性特征。

在图像分辨率方面，图6显示Objects365拥有更大且更多样化的图像分辨率。

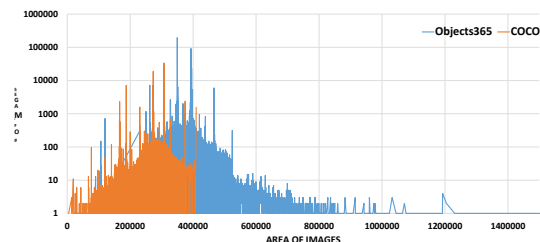


图6. 图像分辨率的分布。

图7提供了“有效标注”与PASCAL VOC和COCO基准的比较结果。通过从Objects365中选取COCO的80个类别（即忽略其他物体类别），我们的数据集在每张图像中拥有更多的边界框（9.04对7.34），以及更多的物体类别数（3.16对2.92）。通过将物体类别映射到VOC的20个类别中，也可以得出类似结论。我们还比较了有效标注面积，即标注物体面积占图像总面积的比例。我们的Objects365提供了明显更多的标注，占比达63%，而COCO和VOC分别为57%和53%。

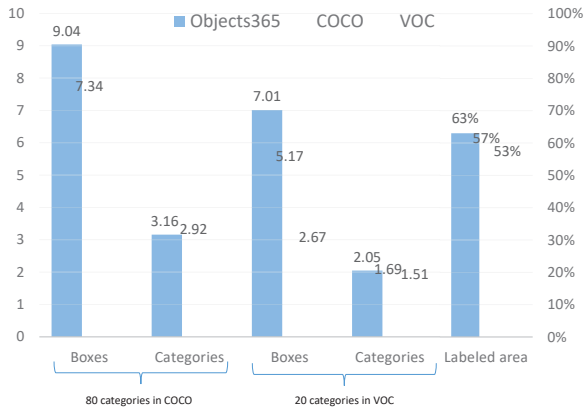


Figure 7. A comparison of the statistics of effective annotations with VOC and COCO datasets.

3.4. Quality

To validate the quality of our Objects365 dataset, three well-trained annotators are asked to label 200 randomly selected images. In total, there are 3250 bounding-boxes based on the refinement from the annotators. 92% of instances are annotated in the original annotations. A comparison of annotation recall with COCO and OpenImages can be found in Table 3.

Table 3. Comparison of annotation recall.

Dataset	OpenImage	COCO	Objects365
Recall(%)	43	83	92

For the annotation precision, we consider a false positive if the object category is wrong or the annotation bounding-box is not accurate. The precision of our Objects365 is obviously higher than COCO with 91.7% vs 71.9%.

4. Experiment

As Objects365 is a new benchmark, we first provide the baseline results based on two widely used detectors, FPN [22] and RetinaNet [23]. Then, we study the generalization ability.

4.1. Experiment Setup

The COCO style mmAP is adopted to evaluate the performance of our Objects365 benchmark. More specifically, we average the IoU=.50:.05:.95 for all the object categories.

For the implementation details, we follow the setting defined in Detectron [14, 15] for COCO. We train our detector on 8 1080-Ti GPUs, with a batch-size of 16. The input image size is 800×1333 , which is also the same in the training process and testing process. A similar setting is utilized for training Objects365 except for the learning rate schedule, where we adopt a learning rate of 0.02, decreased by a factor of 10 at 900K and 1200K, and stopped at 1350K iterations. To speed up the convergence, we apply Syn the Batch

Normalization technique in [26]. For the other datasets, we follow the standard setting.

4.2. Results on Object365

In Table 4, we compare the results of FPN and RetinaNet. The algorithms are trained on the Objects365 training set and the results are reported on the Objects365 validation set as described in Table 2. The mmAP for FPN is 22.5, which is significantly lower than the result on the COCO benchmark (38.3). This shows that our benchmark is more challenging. To analyze the results of our Objects365 dataset, we select the 80 categories defined in COCO from the 365 categories and obtain the mmAP of 38.5 for FPN, 34.5 for RetinaNet, which is comparable to the results on the COCO benchmark. We can see that the low mmAP score on Objects365 is due to the large number of categories in Objects365 dataset, especially those categories which are not defined in COCO dataset.

In addition, we perform diagnosis based on [18] and the results can be found in Figure 8. By comparing the result on Objects365 (left) and the result on COCO (right), we can see that the main gap for the low performance for Objects365 lies on the recall (the gap between BG and FN). A large number of object instances have been missed in our Objects365 dataset. Also, there exist a few object categories which do not have any positive matches, like *radish* and *saw*. In Figure 9, we show three examples for the results on the Objects365. We can see that the false negative may exist in small objects like glove/skis as well as the rare objects like antelope/swing.

4.3. Generalization Ability of Object365

In this sub-section, we study the generalization ability of Object365 as the pretraining dataset for object detection and semantic segmentation. For the detection problem, we select COCO and Pascal VOC to evaluate the general object detection, CityPersons to evaluate the ability on the pedestrian detection. For semantic segmentation, two standard benchmarks like PASCAL VOC and ADE are adopted.

4.3.1 Learning Rate Strategy

We will first discuss how to design the learning rate strategy for finetuning. Let us take the COCO dataset as an example. First, we simply adopt the standard learning rate strategy in [14] for a finetuning experiment. As shown in Table 5, this standard fine-tune strategy based on Objects365 (the third column in Table 5) only provides a small improvement (40.4 vs 39.3) compared with the ImageNet pretraining (shown in the second column of Table 5 with 540K iterations training for the COCO dataset).

To further exploit the finetuning capability of Objects365, we analyze the difference between Objects365

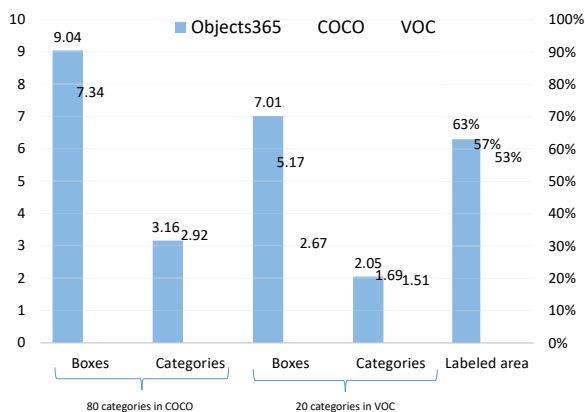


图7. 有效标注统计与VOC和COCO数据集的对比。

3.4. 质量

为了验证我们Objects365数据集的质量，我们邀请了三名训练有素的标注员对200张随机选取的图像进行标注。经过标注员的细化调整，总共生成了3250个边界框。原始标注中92%的实例已获得标注。标注召回率与COCO及OpenImages的对比结果见表3。

表3. 标注召回率对比。

Dataset	OpenImage	COCO	Objects365
Recall(%)	43	83	92

在标注精度方面，如果物体类别错误或标注边界框不准确，我们即视为误报。我们的Objects365数据集精度明显高于COCO数据集，达到91.7%对比71.9%。

4. 实验

由于Objects365是一个新的基准，我们首先基于两种广泛使用的检测器FPN [22]和RetinaNet [23]提供基线结果。然后，我们研究其泛化能力。

4.1. 实验设置

COCO风格的mmAP被用来评估我们Objects365基准的性能。更具体地说，我们平均了所有物体类别的IoU = .50:.05:.95。

在实现细节方面，我们遵循Detectron [14, 15]为COCO数据集定义的设置。我们在8块1080-Ti GPU上训练检测器，批处理大小为16。输入图像尺寸为800×1333，该尺寸在训练和测试过程中保持一致。对于Objects365数据集的训练，除学习率调度策略外，我们采用类似设置：初始学习率为0.02，在90万次和120万次迭代时降至原值的十分之一，并在135万次迭代时终止训练。为加速收敛，我们应用了同步批处理技术。

[26]中的归一化技术。对于其他数据集，我们遵循标准设置。

4.2. Object365数据集上的结果

在表4中，我们比较了FPN和RetinaNet的结果。算法在Objects365训练集上进行训练，结果按照表2所述在Objects365验证集上报告。FPN的mmAP为22.5，显著低于其在COCO基准测试中的结果（38.3）。这表明我们的基准测试更具挑战性。为了分析Objects365数据集的结果，我们从365个类别中选取了COCO定义的80个类别，得到FPN的mmAP为38.5，RetinaNet的mmAP为34.5，这与COCO基准测试的结果相当。可以看出，Objects365上较低的mmAP分数是由于数据集中类别数量庞大，尤其是那些在COCO数据集中未定义的类别。

此外，我们基于[18]进行了诊断，结果如图8所示。通过比较Objects365（左）和COCO（右）的结果，可以看出Objects365性能较低的主要差距在于召回率（背景与假阴性之间的差距）。在我们的Objects365数据集中，大量物体实例被遗漏。此外，存在一些物体类别没有任何正匹配，如radish和saw。在图9中，我们展示了Objects365结果的三个示例。可以看到，假阴性可能出现在小物体（如手套/滑雪板）以及稀有物体（如羚羊/秋千）中。

4.3. Object365的泛化能力

在本小节中，我们研究了Object365作为目标检测和语义分割预训练数据集的泛化能力。对于检测问题，我们选择COCO和Pascal VOC来评估通用目标检测性能，选用CityPersons来评估行人检测能力。在语义分割方面，则采用了PASCAL VOC和ADE这两个标准基准数据集。

4.3.1 学习率策略

我们将首先讨论如何设计微调的学习率策略。以COCO数据集为例，我们初步采用[14]中的标准学习率策略进行微调实验。如表5所示，基于Objects365的这种标准微调策略（表5第三列）相比ImageNet预训练（表5第二列，在COCO数据集上训练540K次迭代）仅带来小幅提升（40.4对比39.3）。

为了进一步挖掘Objects365的微调潜力，我们分析了Objects365之间的差异

Method	$mmAP$	AP^{50}	AP^{75}	AP^S	AP^M	AP^L	AR^1	AR^{10}	AR^{100}	AR^S	AR^M	AR^L
FPN	22.5	35.5	24.3	12.8	24.4	32.8	23.0	36.7	38.1	25.2	40.7	49.8
RetinaNet	18.7	27.3	20.4	9.0	21.1	28.8	21.3	33.3	34.4	19.0	38.4	50.1

Table 4. Results of the baseline algorithms on the Objects365 dataset.

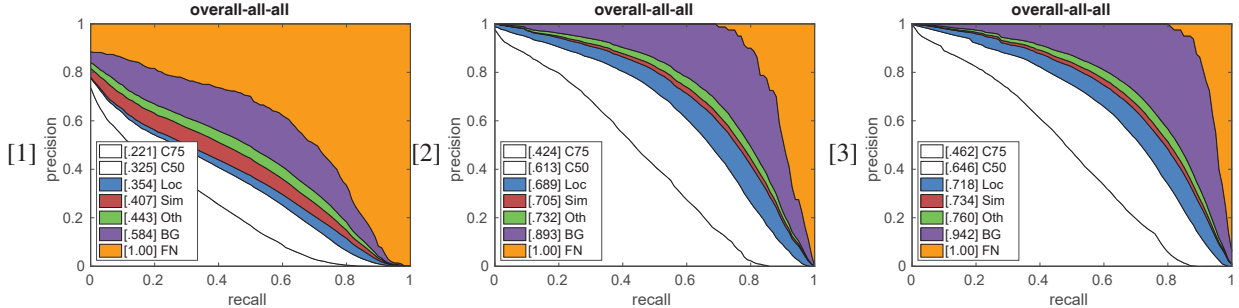


Figure 8. Diagnosis results on Objects365 and COCO. Three figures denote the result on Objects365 (left), the result on COCO with the model pretrained from ImageNet (middle), the result on COCO with the model pretrained from Objects365 (right). All the algorithms are implemented with FPN based on Resnet50 backbone.

Pretraining Dataset	ImageNet	Obj365	Obj365
mmAP	39.3	40.4	42.0
lr@begining	0.02	0.02	0.003
lr@convergence	0.0002	0.0002	0.00012
L2@beginning (1e3)	5.67	0.85	0.85
L2@convergence (1e3)	1.42	1.3	0.85

Table 5. A comparison of different learning rate strategies for fine-tuning on the COCO benchmark. We list the L2-Norm of the weights from the pretrained model of ImageNet and Objects365 at the beginning and convergence of the training. From left to right, these three fine-tuning experiments take 540K, 180K and 90K iterations, respectively.

pre-trained weights and ImageNet pre-trained weights. As shown in the last row of the Table 5, we find that the pre-trained weights from Objects365 have much smaller L2-Norm compared with the ImageNet pre-trained weights. It is about **0.15** times smaller (0.85 vs 5.67) at the beginning of training. Intuitively, the smaller weights usually require a smaller learning rate to train. Therefore, we propose to use a smaller learning rate for Objects365 pre-trained models. Specifically, we design the learning rate according to the ratio of the pre-trained weights’ L2-Norm between Objects365 and ImageNet. At the beginning of the training, we multiply the original learning rate with the L2-norm ratio of 0.15 and set the new learning rate as $3e^{-3} = 0.15 \times 0.02$. We decrease the learning rate by multiplying 1/5 twice during the training process. As Table 5 shows, our new learning rate strategy (the fourth column) can further improve the finetuning capability of the Objects365, which yields a large gain (42.0 vs 40.4).

4.3.2 Finetuning Results

COCO Detection According to Table 6, our Objects365 dataset shows strong generalization ability. Compared with

Method	Pretrain Dataset	Iters	mmAP
FPN	None	540K	39.4
FPN	ImageNet	90K	36.4
FPN	ImageNet	180K	38.3
FPN	ImageNet	540K	39.3
FPN	OpenImages	90k	37.4
FPN	Objects365 w/o COCO 80	90K	39.6
FPN	Objects365	90K	42.0
FPN	ImageNet -> Objects365	90K	42.3
RetinaNet	ImageNet	180K	37.0
RetinaNet	Objects365	180K	39.5
RetinaNet	ImageNet -> Objects365	90K	41.0

Table 6. Generalization ability of general object detection results on the COCO dataset. “Iters” denotes the number of iterations for finetuning the models on the COCO dataset.

FPN and RetinaNet baseline, our finetuned model can significantly boost the performance, with 5.6 points (42 vs 36.4) over the baseline for FPN with the 90K iterations, and 2.5 points (39.5 vs 37) for RetinaNet with the 180K iterations. By pretraining the Objects365 dataset which removes the 80 categories from COCO (Objects365 w/o COCO 80), we can still achieve competitive results with 39.6 on the COCO benchmark. Compared with OpenImages, our Objects365 benchmark has much higher performance gain. By involving ImageNet Pretraining before Objects365, we can slightly improve the FPN with 0.3 (42.3 vs 42). But for the RetinaNet, whose parameters mainly lie on the backbone part, the ImageNet pretraining before Objects365 can further bring in 1.5 points gain (41 vs 39.5).

According to Figure 8 (middle and right figures), the Objects365 pretrained model (right figure) can significantly improve the classification ability of the ImageNet pretrained model (middle figure). The performance gain is mainly due to two factors: the large number of object categories and the

Method	$mmAP$	AP^{50}	AP^{75}	AP^S	AP^M	AP^L	AR^1	AR^{10}	AR^{100}	AR^S	AR^M	AR^L
FPN	22.5	35.5	24.3	12.8	24.4	32.8	23.0	36.7	38.1	25.2	40.7	49.8
RetinaNet	18.7	27.3	20.4	9.0	21.1	28.8	21.3	33.3	34.4	19.0	38.4	50.1

Table 4. Results of the baseline algorithms on the Objects365 dataset.

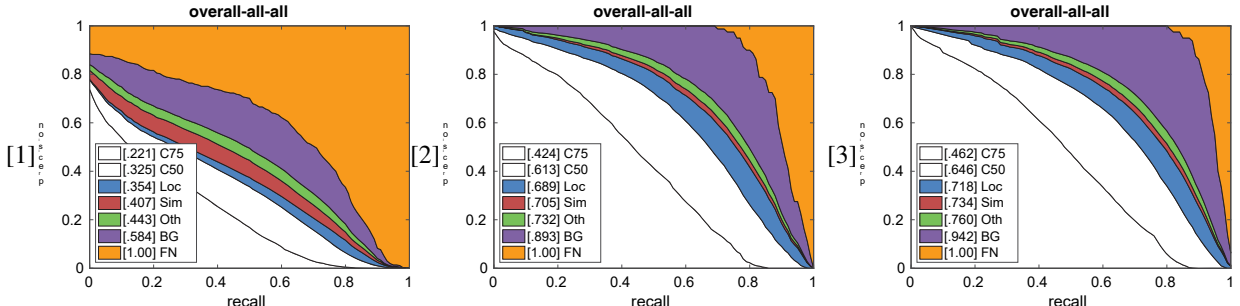


图8. 在Objects365和COCO上的诊断结果。三幅图分别表示在Objects365上的结果（左）、使用基于ImageNet预训练模型在COCO上的结果（中）、使用基于Objects365预训练模型在COCO上的结果（右）。所有算法均采用基于Resnet50骨干网络的FPN实现。

Pretraining Dataset	ImageNet	Obj365	Obj365
mmAP	39.3	40.4	42.0
lr@begining	0.02	0.02	0.003
lr@convergence	0.0002	0.0002	0.00012
L2@beginning (1e3)	5.67	0.85	0.85
L2@convergence (1e3)	1.42	1.3	0.85

表5. 在COCO基准上进行微调时不同学习率策略的比较。我们列出了训练开始和收敛时来自ImageNet和Objects365预训练模型的权重L2范数。从左到右，这三个微调实验分别进行了540K、180K和90K次迭代。

预训练权重和ImageNet预训练权重。如表5最后一行所示，我们发现Objects365的预训练权重相比ImageNet预训练权重具有更小的L2范数。在训练开始时，其L2范数约为ImageNet权重的0.15倍（0.85对比5.67）。直观而言，较小的权重通常需要更小的学习率进行训练。因此，我们建议对Objects365预训练模型采用更小的学习率。具体而言，我们根据Objects365与ImageNet预训练权重的L2范数比值来设计学习率。在训练初始阶段，我们将原始学习率乘以0.15的L2范数比值，将新学习率设定为 $3e^{-3} = 0.15 \times 0.02$ 。在训练过程中，我们通过两次乘以1/5的系数来降低学习率。如表5所示，我们的新的学习率策略（第四列）能进一步提升Objects365的微调能力，带来显著增益（42.0对比40.4）。

4.3.2 微调结果

COCO检测 根据表6，我们的Objects365数据集展现出强大的泛化能力。与

Method	Pretrain Dataset	Iters	mmAP
FPN	None	540K	39.4
FPN	ImageNet	90K	36.4
FPN	ImageNet	180K	38.3
FPN	ImageNet	540K	39.3
FPN	OpenImages	90k	37.4
FPN	Objects365 w/o COCO 80	90K	39.6
FPN	Objects365	90K	42.0
FPN	ImageNet -> Objects365	90K	42.3
RetinaNet	ImageNet	180K	37.0
RetinaNet	Objects365	180K	39.5
RetinaNet	ImageNet -> Objects365	90K	41.0

表6. COCO数据集上通用物体检测结果的泛化能力。“Iters”表示在COCO数据集上微调模型的迭代次数。

与FPN和RetinaNet基线相比，我们的微调模型能显著提升性能：在90K次迭代的FPN上，相比基线提升5.6分（42对36.4）；在180K次迭代的RetinaNet上，提升2.5分（39.5对37）。通过使用移除COCO中80个类别的Objects365数据集进行预训练（Objects365 w/o COCO 80），我们仍能在COCO基准上取得39.6的竞争性结果。与OpenImages相比，我们的Objects365基准带来更显著的性能提升。若在Objects365前加入ImageNet预训练，FPN可微增0.3分（42.3对42）。但对于参数主要集中在骨干网络的RetinaNet，在Objects365前进行ImageNet预训练能额外带来1.5分提升（41对39.5）。

根据图8（中间和右侧图），Objects365预训练模型（右侧图）能显著提升ImageNet预训练模型（中间图）的分类能力。性能提升主要归因于两个因素：大量的物体类别以及

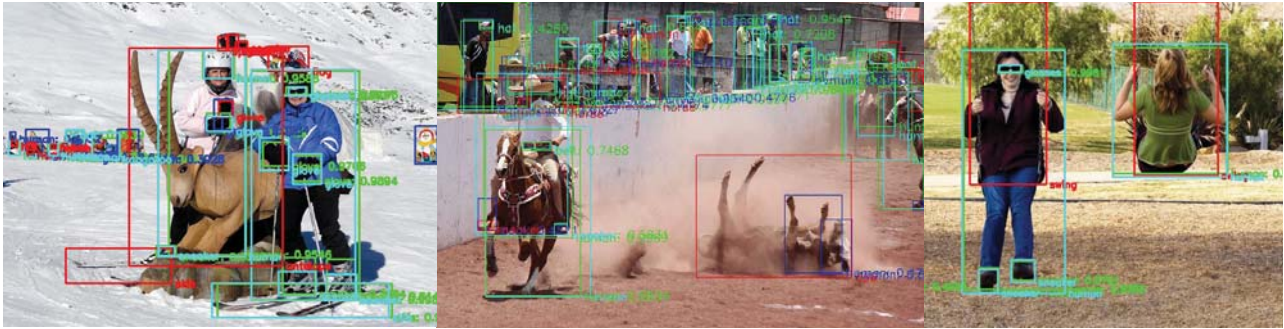


Figure 9. An illustration of the results on the Objects365 dataset. FPN with Resnet50 backbone is utilized. The green bounding-boxes denote the ground-truth (GT) which are positive matched by the predicted bounding-boxes colored with light-blue. Red and blue bounding-boxes denote the false negative GT and false positive predictions, respectively.

Pretraining Dataset	mAP
None	63.4
ImageNet	80.2
ImageNet -> COCO 540K iters	85.1
OpenImages	82.4
Objects365	86.2
ImageNet -> Objects365	86.7

Table 7. Generalization ability of object detection results on the PASCAL VOC dataset. The results are implemented based on FPN with Resnet50 backbone.

improvement of localization ability. According to the gap between “FN” and “Loc”, the Objects365 pretrained model can remarkably improve 2.9 points from 0.689 to 0.718 due to the improvement of the classification ability by involving more object categories. Also, the localization ability is improved by 0.9 if we compare the gap between C75 and Loc.

VOC Detection As shown in Table 7, our Objects365 can provide superior finetuning ability compared with ImageNet and OpenImages. More specifically, we have 6 points gain (86.2 vs 80.2) compared with the ImageNet pre-trained model and more than 1.1 points gain (86.2 vs 85.1) compared with the COCO 540K iterations finetuned model.

CityPersons Besides general object detection, it is also important to validate the generalization ability on the specific object detection problem like pedestrian detection.

In this experiment, we adopt CityPersons [36], one of the standard benchmarks for pedestrian detection. The evaluation follows MR (miss-rate), which is widely used in the pedestrian detection community. We employ FPN as our baseline algorithm. We train the network for 180k iterations.

Table 8 shows the results of pedestrian detection on the CityPersons dataset. We find that pretraining with Objects365 can significantly outperform the baseline algo-

Pretraining Dataset	MR
None	39.49
ImageNet	18.04
ImageNet -> COCO 540K iters	16.24
OpenImages	16.78
Objects365	12.12
ImageNet -> Objects365	12.52

Table 8. Generalization ability of pedestrian detection results on the CityPersons dataset. The results are implemented based on FPN with Resnet50 backbone.

rithms with more than 4 points gain.

VOC Segmentation In addition to the generalization ability for the detection task, we also validate the generalization ability of our Objects365 based on two semantic segmentation benchmarks: PASCAL VOC [8] and ADE [38].

PASCAL VOC contains 20 object categories with pixel-level annotations. The metric of mIOU (mean Intersection-over-Union) is employed to evaluate the performance. PSP-Net [37] with Resnet50 backbone is adopted as the baseline algorithm.

According to Table 9, our algorithm can provide meaningful performance gain by pretraining the backbone part based on Objects365. More specifically, we have more than 2 points over the ImageNet pre-trained (76.7 vs 74.5) and OpenImages pre-trained (76.7 vs 74.1) models, and 1.8 points (76.7 vs 74.9) over the COCO 540K iterations finetuned model.

ADE ADE [38] is a scene parsing dataset introduced in 2017 with around 20K training images from 150 object categories. We report the results on the validation set with 2000 images. According to the Table 10, our Objects365 pretraining model has large advantages over other pretraining models, i.e., 1.7 points (42.9 vs 41.2) over the ImageNet pretraining model and 1.4 points (42.9 vs 41.5) over the COCO 540K iterations finetuned model.

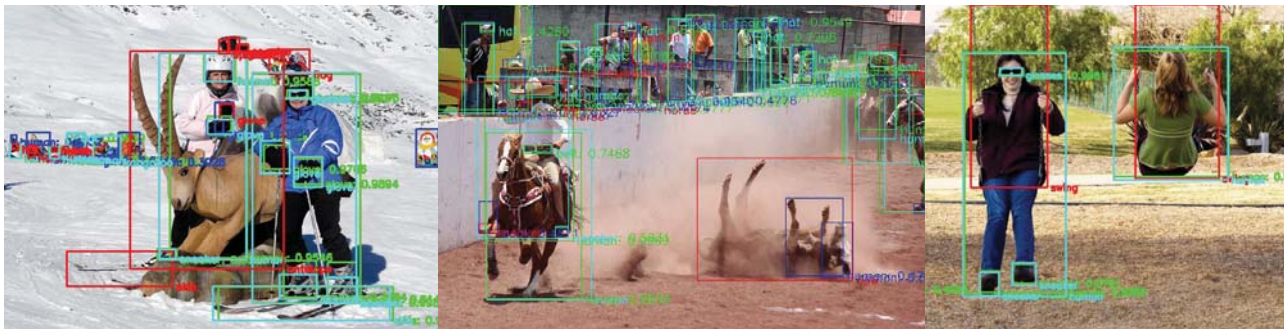


图9. Objects365数据集上的结果示意图。采用以Resnet50为骨干网络的FPN。绿色边界框表示真实标注（GT），它们与浅蓝色预测边界框正匹配。红色和蓝色边界框分别表示假阴性GT和假阳性预测。

Pretraining Dataset	mAP
None	63.4
ImageNet	80.2
ImageNet -> COCO 540K iters	85.1
OpenImages	82.4
Objects365	86.2
ImageNet -> Objects365	86.7

表7. PASCAL VOC数据集上目标检测结果的泛化能力。结果基于Resnet50骨干网络的FPN实现。

定位能力的提升。根据“FN”与“Loc”之间的差距，Objects365预训练模型通过引入更多物体类别提升了分类能力，从而显著提高了2.9个点，从0.689提升至0.718。此外，通过比较C75与Loc之间的差距，定位能力也提升了0.9。

VOC检测 如表7所示，与ImageNet和OpenImages相比，我们的Objects365能够提供更优越的微调能力。具体而言，与ImageNet预训练模型相比，我们获得了6个百分点的提升（86.2对比80.2）；与经过54万次迭代微调的COCO模型相比，则获得了超过1.1个百分点的提升（86.2对比85.1）。

CityPersons 除了通用物体检测外，验证在特定物体检测问题（如行人检测）上的泛化能力也至关重要。

在本实验中，我们采用行人检测领域的标准基准之一CityPersons [36]。评估遵循行人检测社区广泛使用的MR（漏检率）指标。我们采用FPN作为基线算法，并对网络进行了180k次迭代训练。

表8展示了在CityPersons数据集上的行人检测结果。我们发现，使用Objects365进行预训练可以显著超越基线算法。

Pretraining Dataset	MR
None	39.49
ImageNet	18.04
ImageNet -> COCO 540K iters	16.24
OpenImages	16.78
Objects365	12.12
ImageNet -> Objects365	12.52

表8. CityPersons数据集上行人检测结果的泛化能力。结果基于以Resnet50为骨干网络的FPN实现。

超过4点增益的算法。

VOC分割 除了检测任务的泛化能力外，我们还基于两个语义分割基准：PASCAL VOC [8] 和 ADE [38]，验证了基于Objects365的泛化能力。

PASCAL VOC数据集包含20个物体类别，并带有像素级标注。采用mIOU（平均交并比）作为评估性能的指标。以Resnet50为主干网络的PSP-Net [37]被采用为基线算法。

根据表9，我们的算法通过在Objects365上预训练骨干部分能够带来有意义的性能提升。具体而言，相比ImageNet预训练模型（76.7对74.5）和OpenImages预训练模型（76.7对74.1），我们获得了超过2个点的优势；相比COCO 540K迭代微调模型（76.7对74.9），则提升了1.8个点。

ADE ADE [38] 是一个于2017年引入的场景解析数据集，包含约20,000张训练图像，涵盖150个物体类别。我们在包含2000张图像的验证集上报告结果。根据表10所示，我们的Objects365预训练模型相比其他预训练模型具有显著优势，即比ImageNet预训练模型高出1.7个百分点（42.9对比41.2），比COCO 540K迭代微调模型高出1.4个百分点（42.9对比41.5）。

Pretraining Dataset	mIOU
None	58.3
ImageNet	74.5
ImageNet -> COCO 540K iters	74.9
OpenImages	74.1
Objects365	76.7
ImageNet -> Objects365	76.6

Table 9. Generalization ability of semantic segmentation results on the PASCAL VOC dataset. The results are implemented based on PSPNet with Resnet50 backbone.

Pretraining Dataset	mIOU
None	30.2
ImageNet	41.2
ImageNet -> COCO 540K iters	41.5
OpenImages	40.7
Objects365	42.9
ImageNet -> Objects365	43.3

Table 10. Generalization ability of semantic segmentation results on the ADE dataset. The results are implemented based on PSPNet with Resnet50 backbone.

4.3.3 Speed Up Finetuning

According to our experiments on COCO, we find that Objects365 dataset could help researchers to accelerate their finetuning processes. As shown in Table 11, the algorithm trained only 12K iterations based on the Objects365 pre-trained model can have the comparable performance against the model trained with 540K iterations based on the ImageNet pre-trained model. Therefore, by utilizing the Objects365 pretraining, we can obtain more than 10-20 times faster training time without compromising the performance on the COCO benchmark. This significantly reduces the training cost and speed-up the innovation cycle.

4.3.4 Upper-bound

As studied in [15], given sufficient training time, training from scratch can obtain comparable performance as the training from ImageNet pretraining. The model pre-trained with our Objects365 dataset can significantly outperform the model with sufficient long training time, 42.0 vs 39.3 as shown in Table 11. It validates that the Objects365 pre-trained model can further push the upper-bound results for the existing algorithms. There are two potential reasons for the large improvement. First, the feature learned on larger scale dataset is better. Second, the pretraining of both backbone and head provides meaningful gains compared with pretraining the backbone only.

Train Dataset	Iterations	mmAP
ImageNet-> COCO	90K	36.4
ImageNet-> COCO	180K	38.3
ImageNet-> COCO	420K	39.3
ImageNet-> COCO	540K	39.3
Objects365-> COCO	12K	39.0
Objects365-> COCO	18K	40.0
Objects365-> COCO	36K	41.1
Objects365-> COCO	90K	42.0

Table 11. Comparison of the training time for the COCO general detection task. The algorithm is implemented based on the FPN with the Resnet50 backbone. Iterations denotes the number of iterations for the COCO training.

Method	Pretrain Part	Iters	mmAP
ImageNet	Backbone	90K	36.4
ImageNet	Backbone	180K	38.3
ImageNet	Backbone	540K	39.3
Objects365	Backbone	90K	37.8
Objects365	Backbone	180K	39.4
Objects365	Backbone	540K	40.3
Objects365	Backbone & Head	90K	42.0

Table 12. Comparison of the pretraining backbone only against pretraining both the backbone and head on the COCO benchmark. The results are implemented based on FPN with Resnet50 backbone. "Iters" denotes the number of iterations for the COCO training.

4.3.5 Pretrain Backbone vs Pretrain Backbone+Head

One of the main advantages of pretraining with our Objects365 benchmark is due to the pretraining weights of both the backbone and head parts, instead of ImageNet pretraining which only provides the weights for the backbone part. In Table 12, we show an experiment in which by dropping the pretraining weights (randomly initializing the weights) of the head part from a pre-trained Objects365 model, the performance of the finetuned detector drops around 2.6 points (42.0 vs 39.4) on the COCO benchmark. This validates that pretraining head is also important.

5. Conclusion

In this paper, we present a large-scale, high-quality object detection dataset, Objects365, which establishes a new challenge and benefits the many existing localization-sensitive vision tasks. In the future, we plan to investigate bigger models than ResNet-50.

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Pretraining Dataset	mIOU
None	58.3
ImageNet	74.5
ImageNet -> COCO 540K iters	74.9
OpenImages	74.1
Objects365	76.7
ImageNet -> Objects365	76.6

表9. PASCAL VOC数据集上语义分割结果的泛化能力。结果基于以Resnet50为骨干网络的PSPNet实现。

Pretraining Dataset	mIOU
None	30.2
ImageNet	41.2
ImageNet -> COCO 540K iters	41.5
OpenImages	40.7
Objects365	42.9
ImageNet -> Objects365	43.3

表10. ADE数据集上语义分割结果的泛化能力。结果基于PSPNet与Resnet50骨干网络实现。

4.3.3 加速微调

根据我们在COCO上的实验，我们发现Objects365数据集能帮助研究者加速其微调过程。如表11所示，基于Objects365预训练模型仅训练12K次迭代的算法，其性能可与基于ImageNet预训练模型训练540K次迭代的模型相媲美。因此，通过利用Objects365预训练，我们能在不损害COCO基准性能的前提下，获得超过10-20倍的训练速度提升。这显著降低了训练成本，并加速了创新周期。

4.3.4 上界

如[15]所述，在训练时间充足的情况下，从零开始训练可以获得与ImageNet预训练相当的性能。使用我们的Objects365数据集预训练的模型显著优于经过充分长时间训练的模型，如表11所示，结果为42.0对39.3。这验证了Objects365预训练模型能够进一步提升现有算法的上限性能。这一显著改进可能有两个原因：首先，在更大规模数据集上学习到的特征更优；其次，与仅预训练骨干网络相比，同时预训练骨干网络和检测头能带来显著收益。

Train Dataset	Iterations	mmAP
ImageNet-> COCO	90K	36.4
ImageNet-> COCO	180K	38.3
ImageNet-> COCO	420K	39.3
ImageNet-> COCO	540K	39.3
Objects365-> COCO	12K	39.0
Objects365-> COCO	18K	40.0
Objects365-> COCO	36K	41.1
Objects365-> COCO	90K	42.0

表11. COCO通用检测任务的训练时间对比。该算法基于带有ResNet50骨干网络的FPN实现。迭代次数表示COCO训练的迭代轮数。

Method	Pretrain Part	Iters	mmAP
ImageNet	Backbone	90K	36.4
ImageNet	Backbone	180K	38.3
ImageNet	Backbone	540K	39.3
Objects365	Backbone	90K	37.8
Objects365	Backbone	180K	39.4
Objects365	Backbone	540K	40.3
Objects365	Backbone & Head	90K	42.0

表12. 在COCO基准上仅预训练主干网络与同时预训练主干和头部的对比。结果基于采用ResNet50主干的FPN实现。“Iter s”表示COCO训练的迭代次数。

4.3.5 预训练主干网络 vs 预训练主干网络+头部

我们使用Objects365基准进行预训练的主要优势之一在于，其同时提供了主干网络和检测头的预训练权重，而ImageNet预训练仅提供主干网络的权重。在表12中，我们通过实验表明：若从预训练的Objects365模型中移除检测头的预训练权重（随机初始化检测头权重），在COCO基准上微调后的检测器性能会下降约2.6个点（42.0对比39.4）。这验证了检测头预训练同样具有重要意义。

5. 结论

在本文中，我们提出了一个大规模、高质量的物体检测数据集Objects365，它不仅为现有众多对定位敏感的视觉任务带来了新的挑战与助益，也为相关研究设立了新的标杆。未来，我们计划探索比ResNet-50更庞大的模型架构。

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